

## GEMM Analytical Model - Equations

Auto-generated from PYTHON\_FORMULAS and LATEX\_SYMBOL\_OVERRIDES in gemmanalytics.py.

Output order: execution

**row 22 - INPUT\_A\_BYTES\_PER\_ELEMENT**

$$\text{Bytes}_{\text{pElement}}^{(\text{matA})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matA})}]$$

**row 23 - INPUT\_B\_BYTES\_PER\_ELEMENT**

$$\text{Bytes}_{\text{pElement}}^{(\text{matB})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matB})}]$$

**row 25 - OUTPUT\_BYTES\_PER\_ELEMENT\_AFTER\_DOWN\_CONVERSION**

$$\text{Bytes}_{\text{pElement}}^{(\text{D}\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matD}\downarrow)}]$$

**row 33 - EU\_COUNT**

$$|\text{EU}| = |\text{XeCore}|_{\text{pXeCU}} \times |\text{EU}|_{\text{pXeCore}} \times |\text{XeCU}|$$

**row 102 - TOTAL\_L2\_SIZE\_B\_FOR\_A\_SINGLE\_INSTANCE**

$$|\text{L2Bytes}| = \text{L2BankSize}_{\text{MB}} \times |\text{L2Banks}|_{\text{pXeCU}} \times 1024 \times 1024$$

**row 125 - MAX\_POSSIBLE\_HBM\_BW\_FREQ\_B\_CLK**

$$\text{MemBW}_{\text{BpClk}}^{\text{max}} = \frac{\text{MemBW}_{\text{GBps}}^{\text{max}}}{f_{\text{GHz}}^{(\text{GT})}}$$

**row 36 - MMA\_MAC\_THROUGHPUT\_PER\_XECORE**

$$\tau_{\text{mMACpClk} \cdot \text{XeCore}}^{(\text{peak})} = \min\left(\frac{4}{\frac{\text{FLOOR}(\text{Bytes}_{\text{pElement}}^{(\text{matA})} \times 2, 1)}{2}}, \frac{4}{\frac{\text{FLOOR}(\text{Bytes}_{\text{pElement}}^{(\text{matB})} \times 2, 1)}{2}}\right) \times \text{D}_{\text{DPAS}} \times 16 \times |\text{EU}|_{\text{pXeCore}}$$

**row 42 - CLKS\_PER\_DPAS**

$$\text{CLKS}_{\text{DPAS}} = \frac{M_{\text{pThread}} \times K_{\text{pThread}} \times N_{\text{pThread}}}{\frac{T_{\text{mMACpClk-XeCore}}^{(\text{peak})}}{|EU|_{\text{pXeCore}}}}$$

**row 13 - WAVES**

$$N_{\text{waves}} = \text{CEILING}(|\text{Tiles}|_{\text{N}}^{(\text{GPU})}, 1) \times \text{CEILING}(|\text{Tiles}|_{\text{M}}^{(\text{GPU})}, 1)$$

**row 14 - TG\_TILES\_IN\_N**

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}$$

**row 15 - TG\_TILES\_IN\_M**

$$|\text{Tiles}|_{\text{M}}^{(\text{TG})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{ThreadGroup})}}$$

**row 16 - TG\_CLUSTER\_TILES\_IN\_N**

$$|\text{Tiles}|_{\text{N}}^{(\text{TG Cluster})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCoreCluster})}}$$

**row 17 - TG\_CLUSTER\_TILES\_IN\_M**

$$|\text{Tiles}|_{\text{M}}^{(\text{TG Cluster})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCoreCluster})}}$$

**row 18 - XECU\_TILES\_IN\_N**

$$|\text{Tiles}|_{\text{N}}^{(\text{XeCU})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}$$

**row 19 - XECU\_TILES\_IN\_M**

$$|\text{Tiles}|_M^{(\text{XeCU})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}$$

**row 20 - GPU\_TILES\_IN\_N**

$$|\text{Tiles}|_N^{(\text{GPU})} = \frac{N_{\text{dim}}}{\frac{W_{\text{element}}^{(\text{XeCUTile})}}{W_{\text{XeCU}}^{(\text{GPURTile})}}}$$

**row 21 - GPU\_TILES\_IN\_M**

$$|\text{Tiles}|_M^{(\text{GPU})} = \frac{\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}}{H_{\text{XeCU}}^{(\text{GPURTile})}}$$

**row 43 - THREAD\_WIDTH\_IN\_UNITS\_OF\_ELEMENTS**

$$W_{\text{element}}^{(\text{Thread})} = N_{\text{pThread}}$$

**row 44 - THREAD\_HEIGHT\_IN\_UNITS\_OF\_ELEMENTS**

$$H_{\text{element}}^{(\text{Thread})} = M_{\text{pThread}}$$

**row 47 - TG\_WIDTH\_IN\_UNITS\_OF\_ELEMENT\_REALIZED\_BY\_MULTIPLE\_MMA\_ITERATIONS**

$$W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})} = W_{\text{thread}}^{(\text{ThreadGroup})} \times W_{\text{element}}^{(\text{Thread})}$$

**row 48 - TG\_HEIGHT\_IN\_UNITS\_OF\_ELEMENT**

$$H_{\text{element}}^{(\text{ThreadGroup})} = H_{\text{thread}}^{(\text{ThreadGroup})} \times H_{\text{element}}^{(\text{Thread})}$$

**row 52 - XECORE\_CLUSTER\_WIDTH\_IN\_UNITS\_OF\_ELEMENT**

$$W_{\text{element}}^{(\text{XeCoreCluster})} = W_{\text{TG, keep cluster sizes as 4}}^{(\text{XeCoreCluster})} \times W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

**row 53 - XECORE\_CLUSTER\_HEIGHT\_IN\_UNITS\_OF\_ELEMENT**

$$H_{\text{element}}^{(\text{XeCoreCluster})} = H_{\text{TG,keep cluster sizes4}}^{(\text{XeCoreCluster})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

**row 56 - XECU\_TILE\_WIDTH\_IN\_UNITS\_OF\_ELEMENT**

$$W_{\text{element}}^{(\text{XeCUTile})} = W_{\text{TG}}^{(\text{XECUTile})} \times W_{\text{element}}^{(\text{ThreadGroup,realized by multiple MMA iterations})}$$

**row 57 - XECU\_TILE\_HEIGHT\_IN\_UNITS\_OF\_ELEMENT**

$$H_{\text{element}}^{(\text{XeCUTile})} = H_{\text{TG}}^{(\text{XECUTile})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

**row 59 - GPU\_TILE\_HEIGHT\_IN\_XECU\_UINT**

$$H_{\text{XeCU}}^{(\text{GPUTile})} = \frac{|\text{XeCU}|}{W_{\text{XeCU}}^{(\text{GPUTile})}}$$

**row 60 - GPU\_TILE\_WIDTH\_IN\_UNITS\_OF\_ELEMETNS**

$$W_{\text{element}}^{(\text{GPUTile})} = W_{\text{XeCU}}^{(\text{GPUTile})} \times W_{\text{element}}^{(\text{XeCUTile})}$$

**row 61 - GPU\_TILE\_HEIGHT\_IN\_UNITS\_OF\_ELEMETNS**

$$H_{\text{element}}^{(\text{GPUTile})} = H_{\text{XeCU}}^{(\text{GPUTile})} \times H_{\text{element}}^{(\text{XeCUTile})}$$

**row 63 - MAT\_A\_INPUT\_SIZE\_B**

$$\text{Size}_B^{(\text{matA})} = M_{\text{dim}} \times K_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{matA})}$$

**row 64 - MAT\_B\_INPUT\_SIZE\_B**

$$\text{Size}_B^{(\text{matB})} = K_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{matB})}$$

**row 65 - MAT\_C\_INPUT\_D\_OUTPUT\_SIZE\_B**

$$\text{Size}_{\text{matB}}^{(\text{matC,matD})} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{D}\downarrow)}$$

**row 66 - MAT\_D\_INTERMEDIATE\_SIZE\_B**

$$\text{Size}_B^{(\text{matD}\downarrow)} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Byte}_{\text{pElement}}^{(\text{matD})}$$

**row 103 - WORKING\_DATA\_SET\_SIZE\_OF\_K\_IN\_L2\_CORRESP\_20K\_CLOCKS\_OF\_THREAD\_DIVERGENCE**

$$|\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} = \min(20000 \times \frac{K_{\text{pThread}}}{\text{CLKS}_{\text{DPAS}}}, K_{\text{dim}})$$

**row 104 - TOTAL\_REQUIRED\_L2\_SIZE\_FOR\_IDEAL\_HIT\_RATE\_B\_FOR\_A\_SINGLE\_INSTANCE\_AND\_SINGLE\_WAVE**

$$\text{ance, 1 wave alwayshit} = H_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{pElement}}^{(\text{matA})} \times |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} + W_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{pElement}}^{(\text{matB})} \times |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} + W_{\text{element}}^{(\text{XeCUTile})} \times H_{\text{element}}^{(\text{XeCUTile})}$$

**row 37 - CLK\_SPECIFIED\_EFFICIENCY**

$$T_{\text{clk}}^{(\text{total})} = \frac{\frac{\frac{M_{\text{dim}} \times K_{\text{dim}} \times N_{\text{dim}}}{|\text{XeCore}|_{\text{pXeCU}}}}{|\text{XeCU}|}}{\tau_{\text{mMACpClk-XeCore}}^{(\text{peak})}} \eta_{\text{systolic}}$$

**row 69 - TOTAL\_L2\_READ\_B**

$$\text{L2Rd}_B^{(\text{total})} = \text{Size}_B^{(\text{matA})} \times \text{CEILING}(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}, 1) + \text{Size}_B^{(\text{matB})} \times \text{CEILING}(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{ThreadGroup})}}, 1)$$

**row 70 - TOTAL\_L2\_WRITE\_B**

$$\text{L2Wr}_B^{(\text{total})} = \text{Size}_{\text{matB}}^{(\text{matC, matD})}$$

**row 71 - TOTAL\_L1\_READ\_B**

$$\text{L1Rd}_B^{(\text{total})} = \text{Size}_B^{(\text{matA})} \times \text{CEILING}(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{Thread})}}, 1) + \text{Size}_B^{(\text{matB})} \times \text{CEILING}(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{Thread})}}, 1)$$

**row 72 - TOTAL\_L1\_WRITE\_B**

$$\text{L1Wr}_B^{(\text{total})} = \text{Size}_{\text{matB}}^{(\text{matC, matD})}$$

**row 74 - L2\_READ\_B\_XECORE\_CLK**

$$L2RdB_{p(Cl k \cdot XeCore)} = \frac{\frac{L2Rd_B^{(total)}}{|XeCU| \times |XeCore|_{pXeCU}}}{T_{clk}^{(total)}}$$

**row 75 - L2\_WRITE\_B\_XECORE\_CLK**

$$L2WrB_{p(Cl k \cdot XeCore)} = \frac{\frac{L2Wr_B^{(total)}}{|XeCU| \times |XeCore|_{pXeCU}}}{T_{clk}^{(total)}}$$

**row 76 - L2\_READ\_WRITE\_B\_XECORE\_CLK**

$$L2RdWrB_{p(Cl k \cdot XeCore)} = L2RdB_{p(Cl k \cdot XeCore)} + L2WrB_{p(Cl k \cdot XeCore)}$$

**row 77 - L1\_READ\_B\_XECORE\_CLK**

$$L1RdB_{p(Cl k \cdot XeCore)} = L1RdB_{p(Cl k \cdot EU)} \times |EU|_{pXeCore}$$

**row 78 - L1\_WRITE\_B\_XECORECLK**

$$L1WrB_{p(Cl k \cdot XeCore)} = L1WrB_{p(Cl k \cdot EU)} \times |EU|_{pXeCore}$$

**row 79 - L1\_READ\_B\_EU\_CLK**

$$L1RdB_{p(Cl k \cdot EU)} = \frac{\frac{L1Rd_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

**row 80 - L1\_WRITE\_B\_EU\_CLK**

$$L1WrB_{p(Cl k \cdot EU)} = \frac{\frac{L1Wr_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

**row 83 - L2\_READ\_MAX\_B\_XECORE\_CLK**

$$L2RdB_{p(Cl k \cdot XeCore)}^{(max)} = \frac{|XeCore|_{pXeCU} \times 64}{|XeCore|_{pXeCU}}$$

**row 84 - L2\_WRITE\_MAX\_B\_XECORE\_CLK**

$$L2WrB_{p(Cl k \cdot XeCore)}^{(max)} = L2RdB_{p(Cl k \cdot XeCore)}^{(max)}$$

**row 85 - L2\_READ\_WRITE\_MAX\_B\_XECORE\_CLK**

$$L2RdWrB_{p(Cl k \cdot XeCore)}^{(max)} = L2WrB_{p(Cl k \cdot XeCore)}^{(max)}$$

**row 90 - L2\_READ\_B\_XECORE\_CLK\_PCT**

$$\eta_{L2RdB/p(Cl k \cdot XeCore)} = \frac{L2RdB_{p(Cl k \cdot XeCore)}}{L2RdB_{p(Cl k \cdot XeCore)}^{(max)}}$$

**row 91 - L2\_WRITE\_B\_XECORE\_CLK\_PCT**

$$\eta_{L2WrB/p(Cl k \cdot XeCore)} = \frac{L2WrB_{p(Cl k \cdot XeCore)}}{L2WrB_{p(Cl k \cdot XeCore)}^{(max)}}$$

**row 92 - L2\_READ\_WRITE\_B\_XECORE\_CLK\_PCT**

$$\eta_{L2RdWrB/p(Cl k \cdot XeCore)} = \frac{L2RdWrB_{p(Cl k \cdot XeCore)}}{L2RdWrB_{p(Cl k \cdot XeCore)}^{(max)}}$$

**row 93 - L1\_READ\_B\_EU\_CLK\_PCT**

$$\eta_{L1RdB/p(Cl k \cdot EU)} = \frac{L1RdB_{p(Cl k \cdot EU)}}{L1RdBW_{BpCl k \cdot EU}^{max}}$$

**row 94 - L1\_WRITE\_B\_EU\_CLK\_PCT**

$$\eta_{L1WrB/p(Cl k \cdot EU)} = \frac{L1WrB_{p(Cl k \cdot EU)}}{L1WrBW_{BpCl k \cdot EU}^{\max}}$$

**row 106 - L2\_HIT\_RATE\_ASSUMED\_RANDOM\_ACCESS\_WITHIN\_THE\_WORKING\_DATA\_SET\_PCT**

$$L2HitRate_{\text{random WS access}} = \min\left(\frac{|L2Bytes|}{\text{TotalRequiredL2Size}_{\text{matB alwayshit}}^{1 \text{ instance, 1 wave}}}, 1\right)$$

**row 107 - L2\_MISS\_RATE\_PCT**

$$L2MissRate = 1 - L2HitRate_{\text{random WS access}}$$

**row 108 - TOTAL\_L2\_READ\_TRAFFIC\_B**

$$L2RdB_{\text{total}} = L2RdB_{\text{B}}^{(\text{total})}$$

**row 110 - PROBABILITY\_OF\_MATA\_HIT\_IN\_L2\_DURING\_A\_NON\_FIRST\_WAVE\_PCT**

$$_{\text{1stwave}} = \text{IF}(\text{Size}_{\text{B}}^{(\text{matA})} + \text{Size}_{\text{B}}^{(\text{matB})} + \text{Size}_{\text{matB}}^{(\text{matC}, \text{matD})} \leq |L2Bytes|, 1.0, \text{IF}(\text{K}_{\text{dim}} > 2 \times |WorkingSet|_{20Kclksofthreaddivergence}^{(\text{KinL2})}, 0.0, 1 - \frac{\text{K}_{\text{dim}} - |WorkingSet|_{20Kclksofthreaddivergence}^{(\text{KinL2})}}{|WorkingSet|_{20Kclksofthreaddivergence}^{(\text{KinL2})}})$$

**row 111 - PROBABILITY\_OF\_MATB\_HIT\_IN\_L2\_DURING\_A\_NON\_FIRST\_WAVE\_PCT**

$$P_{L2Hit, \text{after1stwave}}^{(\text{matB})} = \text{IF}(\text{Size}_{\text{B}}^{(\text{matA})} + \text{Size}_{\text{B}}^{(\text{matB})} + \text{Size}_{\text{matB}}^{(\text{matC}, \text{matD})} \leq |L2Bytes|, 1.0, 0.0)$$

**row 112 - PROBABILITY\_OF\_MATA\_MISS\_IN\_L2\_DURING\_A\_NON\_FIRST\_WAVE\_PCT**

$$P_{L2Miss, \text{after1stwave}}^{(\text{matA})} = 1 - P_{L2Hit, \text{after1stwave}}^{(\text{matA})}$$

**row 113 - PROBABILITY\_OF\_MATB\_MISS\_IN\_L2\_DURING\_A\_NON\_FIRST\_WAVE\_PCT**

$$P_{L2Miss, \text{after1stwave}}^{(\text{matB})} = 1 - P_{L2Hit, \text{after1stwave}}^{(\text{matB})}$$

**row 116 - TOTAL\_HBM\_READ\_B\_AFTER\_A\_COMPLETION\_OF\_A\_WAVE\_CONSIDER\_COLD\_CACHE**

$$e_B^{(\text{matB})} + \text{Size}_B^{(\text{matB})} \times \left( \text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}(\text{XeCUTile})}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1stwave}}^{(\text{matB})} + \left( \text{L2RdB}_{\text{total}} - \left( \text{Size}_B^{(\text{matA})} + \text{Size}_B^{(\text{matA})} \times \left( \text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}(\text{XeCUTile})}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1}}^{(\text{matA})} \right) \right) \times 2$$

**row 117 - TOTAL\_HBM\_WRITE\_B**

$$\text{MemWrB}^{(\text{total})} = \text{Size}_{\text{matB}}^{(\text{matC,matD})}$$

**row 118 - TOTAL\_HBM\_B**

$$\text{MemRdWrB}^{(\text{total})} = \text{MemWrB}^{(\text{total})} + \text{MemRdB}_{\text{coldstart}}^{(\text{wave})}$$

**row 121 - HBM\_READ\_B\_CLK**

$$\text{MemRdB}_{\text{pClk}} = \frac{\text{MemRdB}_{\text{coldstart}}^{(\text{wave})}}{T_{\text{clk}}^{(\text{total})}}$$

**row 122 - HBM\_WRITE\_B\_CLK**

$$\text{MemWrB}_{\text{pClk}} = \frac{\text{MemWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

**row 123 - HBM\_TOTAL\_B\_CLK**

$$\text{MemRdWrB}_{\text{pClk}} = \frac{\text{MemRdWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

**row 126 - HBM\_BW\_PCT**

$$\eta_{\text{MemBW}} = \frac{\text{MemRdWrB}_{\text{pClk}}}{\text{MemBW}_{\text{BpClk}}^{\text{max}}}$$

**row 127 - GTI\_READ\_BW\_PCT**

$$\eta_{\text{GTIRdBW}} = \frac{\text{MemRdB}_{\text{pClk}}}{\text{GtiRdBW}_{\text{BpClk}}^{\text{max}}}$$

**row 128 - GTI\_WRITE\_BW\_PCT**

$$\eta_{\text{GTIW_rBW}} = \frac{\text{MemWrB}_{\text{pClk}}}{\text{GtiWrBW}_{\text{BpClk}}^{\text{max}}}$$